

USRSE 2026



Advancing Science in the Age of AI

San Jose, CA · October 19–21, 2026



Full Program

<https://us-rse.org/usrse26/program/>



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Welcome!

Welcome to USRSE'26, the fourth annual US Research Software Engineering conference, hosted by the US Research Software Engineer Association (US-RSE). We are thrilled to have you in San Jose. With this year's theme, "Advancing Science in the Age of AI," a practical question arises: how can research software engineers and their collaborators increase research impact during these rapidly changing times? AI is changing how software gets written, tested, and maintained. This is far from the whole picture, as it brings real hurdles, from code that is easier to generate than to understand, to new questions about correctness, credit, security, and sustainability. These tools should assist our work, not replace the people who do it. The craft of building reliable, reproducible software, and of supporting the people behind it, matters as much now as ever.

We're thrilled about our program. With so many great sessions, choosing among them can be difficult. Across our parallel tracks you will find papers, talks, notebooks, workshops, Birds of a Feather sessions, posters, and our Rapid Access Microtalks, as well as multiple student and early career sessions and our inaugural job fair. AI topics run through much of the program, from retrieval assistants and agentic code modernization to guardrails and trust for AI-assisted development. Just as present are the foundations of our field: reproducible workflows, testing and sustainability, refactoring long-lived code, training and mentorship, user experience, HPC and cross-facility infrastructure, security, and honest reflections on RSE roles and careers. Whatever brought you here, you will find people who share your interests!

US-RSE is built and run by volunteers, and this conference exists only because so many of you gave your time to make it happen. Our planning committee, technical program committee and reviewers, keynote speakers, session chairs, panelists, mentors, early-career organizers, presenters, and on-site volunteers all deserve your thanks. When you meet them this week, please tell them so, and if you would like to help shape what comes next, reach out to a working group chair or a Steering Committee member. We are equally grateful to our sponsors, whose support underpins so much of what happens here and signals that research software, and the engineers who create it, are worth sustaining. Please take time to talk with their representatives during breaks and in the foyer.

This year we hope that you meet someone new, leave with an idea you can use, and return to your work with renewed energy that strengthens the RSE community for years to come.

Alex Koufos and Keith Beattie
General Chairs, USRSE'26

All conference participants are expected to abide by the Code of Conduct. Please report problems or concerns to coc@us-rse.org.

<https://us-rse.org/usrse26/about/code-of-conduct/>

Full program details and conference policies: <https://us-rse.org/usrse26/>

Schedule Overview

Full abstracts and presenter names are in the online program: <https://us-rse.org/usrse26/program/>

Monday, October 19

7:30–8:30am	Registration , <i>Prefunction</i>
8:30–10am	Welcome and Plenary Session , <i>San Jose Ballroom Salon 3 & 4</i> Keynote — Fernando Pérez
10–10:30am	Morning Break , <i>San Jose Marriott</i> US-RSE Office Hours
10:30–11:30am	Panel , <i>San Jose Ballroom Salon 3 & 4</i> Student & Early Career — Careers in RSE: Experiences from Research Software Engineers Across Industry, Academia, and Laboratories
10:30am–12pm	RAG Assistants , <i>San Jose Ballroom Salon 1 & 2</i> • AquiLLM: Evaluating Faithfulness in Open-Weight RAG-LLM Systems for Scientific Research • LZ Oracle: an AI-assisted knowledge retrieval platform for the LUX-ZEPLIN dark matter experiment • Data Sherpa: An AI-Powered Assistant for Scientific Collaboration Knowledge Management • AskALCF: A RAG Assistant for HPC User Support Case Studies Demonstrating RSE Impacts Part I , <i>San Jose Ballroom Salon 5 & 6</i> • Automating Data-Driven Alternative Text in Quarto-based NOAA Fisheries Reports • A Modular Threaded Software Application for Axion Detection with DMRadio • Poisonous Books: Rule Based Physics Pipeline and Agentic LLMs for Detecting and Analyzing Toxic Pigments in Victorian Bookbindings • Building Modern Tools for Computational Science in Flutter to Track the American Chestnut AI Assisted Code Development , <i>Willow Glen Room</i> • Reuse, Compose, Extend, Standardize, Automate: Two Decades of RSEing Open (Neuro)Science at CON • Research Software Engineering at SNS and HFIR: Advancing Neutron Science Software in the Age of AI-Assisted Development • Designing for Trust in AI-Assisted Scientific Computing
12–1:30pm	Lunch Break , <i>San Jose Marriott</i>
1:30–3pm	RSE Team Operations , <i>San Jose Ballroom Salon 3 & 4</i> Birds of a Feather — Navigating RSE Team Operations: Leadership, Sustainability, and Emerging Challenges Institutional RSE Community , <i>San Jose Ballroom Salon 1 & 2</i> Birds of a Feather — Building an Institutional RSE Community Better Scientific Software Fellowship , <i>San Jose Ballroom Salon 5 & 6</i> Birds of a Feather — Better Scientific Software Fellowship Community Threat Modeling for Research Software Systems , <i>Willow Glen Room</i> Workshop — A Group Attempt at Threat Modeling for Research Software Systems
3–3:30pm	Afternoon Break , <i>San Jose Marriott</i>

Monday, October 19, continued

3:30–5pm	Interactive Computing and Reproducible Workflows, <i>San Jose Ballroom Salon 1 & 2</i> • Effective Tutorial Notebooks for Cloud Computing Workflows using Remote Sensing Datasets • Marimo: an open-source reactive notebook for Python • Beyond human readers: designing domain-specific language notebooks for AI-assisted code generation • From Notebook to Control Room: Reproducible MLOps for Accelerator Digital Twins Lessons from Training Users & RSEs, <i>San Jose Ballroom Salon 5 & 6</i> • INTERSECT After Three Years: Lessons Learned from Delivering Advanced Research Software Engineering Bootcamps • Beyond Workshops: RSE-Led Hackathons as Experiential Learning for Software Practice in the AI Era • Building Open Source Communities Through Effective Tutorials • Evolving HPC User Support: Just-in-Time Learning with Instructional Videos AI + Software Quality, <i>Willow Glen Room</i> • Software Sustainability Toolkit: Measuring and Improving Sustainable Research Software at Scale • Measuring What Matters in the Age of AI: The Research Software Metrics Landscape • Back to Basics: Design-First Development in the Age of AI Coding Assistants • Scientific Open-Source Software Is Less Likely to Become Abandoned Than One Might Think: Lessons from Curating a Catalog of Maintained Scientific Software
5:30–7:30pm	Poster Reception, <i>San Jose Ballroom Salon 3 & 4</i>

Tuesday, October 20

7:30–8:30am	Registration, <i>Prefunction</i>
8:30–10am	Plenary II, <i>San Jose Ballroom Salon 3 & 4</i> A panel of experts in position of insight, from national labs, universities and industry, to have a meaningful discussion on the RSE role, it's current state, what the future looks like, and how we may shape that future.
10–10:30am	Morning Break, <i>San Jose Marriott</i> US-RSE Office Hours
10am–3:30pm	Job Fair, <i>Market Street</i>
10:30–11:30am	Making AI Sustainable: Token Economics, <i>San Jose Ballroom Salon 3 & 4</i> • Agents on a Ramen Budget: Token-Smart Agentic Workflows with Local Models • Tokens Are the New Compute Hours: Managing Commercial AI Costs on a Student Budget
10:30am–12pm	RSE Practices in the Age of AI Part I, <i>San Jose Ballroom Salon 1 & 2</i> • Research Software Engineering in the Age of Generative AI • Beyond Editing Code: Uses for AI Tools in Scientific Work • Ten Projects, Two Teams, One Question: How Does AI Impact Development? • Coding is Easy, Everything else is Hard: Lessons learned from working with third-party software developers The Human Factor, <i>San Jose Ballroom Salon 5 & 6</i> • Usability Challenges across AI-Assisted Infrastructure, Security, and Scientific Workflows • From Mood Boards to Web Components: Our Journey Building a Design System for Research • Human in the Loop: Requirements Gathering and Researcher Trust in the Age of AI • Designing for Two Audiences: AI-Readiness and Human Data Usability

Tuesday, October 20, continued

10:30am–12pm <i>continued</i>	HPC + AI, Willow Glen Room • DART: Distributed Assignment of Research Tasks for Heterogeneous Compute Environments • Ten Practices for Refactoring Bioinformatics Pipelines to nf-core DSL2 on Shared Slurm HPC • Tapis AI Assistant: Increasing Research Impact through Traceable and Evaluated Agentic Tools • Structuring agentic AI for HPC code modernization
12–1:30pm	Lunch Break, San Jose Marriott Working Group Fair
12:45–1:15pm	Working Group Fair (Presentations during lunch), San Jose Ballroom Salon 3 & 4
1:30–3pm	RSE Skill Training, San Jose Ballroom Salon 3 & 4 Birds of a Feather — How to train an RSE: A discussion of RSE skill training - past, future, and present User Experience: Journey Mapping, San Jose Ballroom Salon 1 & 2 Workshop — Journey Mapping: A technique to see opportunities and recognize gaps in user experience Improve Research Outputs by Improving Institutional Collaboration, San Jose Ballroom Salon 5 & 6 Birds of a Feather — How can institutional RSE hubs and OSPOs work together for the betterment of research outputs? Refactoring Code for Reuse and Impact, Willow Glen Room Workshop — Fixer-Upper Software: Refactoring Code for Reuse and Impact
3–3:30pm	Afternoon Break, San Jose Marriott
3:30–5pm	Case Studies Demonstrating RSE Impacts Part II, San Jose Ballroom Salon 1 & 2 • Josh: Efficient, Portable, and Scalable Cross-Disciplinary Vegetation Modeling via Domain-Specific Language • Take it on the road: How RCD Helps Field Researchers Report Culvert Condition even when Offline • Escaping the Enclave: Format-Faithful Synthetic Data for Open, Reproducible Medicare Claims Pipelines • Automated Data Reduction for Neutron Scattering Experiments: Current Capabilities and Future Directions Code Modernization Using AI, San Jose Ballroom Salon 5 & 6 • Agentic code modernization • Scientific Agentic Engineering: A Framework for Reliable AI Agents in Research Software • From Laptop to Cluster: HPC Translation Patterns for AI Stacks • Beyond Code Pairs: Generating Test-Grounded Data for AI-Assisted Scientific Software Modernization Workflows, Willow Glen Room • Cloud-Native Graphical and Programmatic Cross-Database Access to Experiment Metadata with Trino, Superset and Ibis • Custom Containers for Continuous Integration via Spack: Decoupling Environment Configuration from Test Execution • OSC-IS: Research Software Infrastructure for Scientific Artifact Preservation, Provenance, and Reproducible Collaboration • Fabla: An Open-Source Voice-First EMA Platform for Clinical Research
6–8pm	Dinner, San Jose Marriott

Wednesday, October 21

7:30–8:30am	Registration , <i>Prefunction</i>
8:30–10am	Plenary III & Sponsor Talks , <i>San Jose Ballroom Salon 3 & 4</i> Announcements, US-RSE Updates, US-RSE Awards, BSSw Fellowship Awards
10–10:30am	Morning Break , <i>San Jose Marriott</i> US-RSE Office Hours
10:30am–12pm	Collaboration & Community Building , <i>San Jose Ballroom Salon 1 & 2</i> • Building Inclusive Talent Pathways: Scaling Outreach, Workforce Development, and Community Engagement for RSE • HPC Carpentry and Community Building at NIST • Navigating the Code Review Process as a Team: A Case Study • Collaboration within RSE group at Princeton: development of slurmise for resource usage estimation Software Quality , <i>San Jose Ballroom Salon 5 & 6</i> • Developing a Handbook of best practices for research software sustainability at Princeton • We Create Quality: Towards a Human-Centric Theory of Research Software Quality in the Age of AI • Beyond Productivity: Human-Centered Guardrails for AI-Assisted Research Software Workflows • Performance Portable Software Design Patterns – What they can do and why we should collect them. Generative AI + Applications , <i>Willow Glen Room</i> • From Chaos to Schema: LLM-Powered Document Information Extraction • Steering an LLM AutoResearch Loop with Domain Context: A Case Study with Vehicle Energy Models • Logic Meets Intuition: The Neuro-Symbolic Upgrade for LLMs
12–1:30pm	Lunch Break , <i>San Jose Marriott</i>
1:30–3pm	RSE Practices in the Age of AI Part II , <i>San Jose Ballroom Salon 1 & 2</i> • AI-Augmented Research Software Engineering: Risks, Challenges, and Practices • A Framework for Evaluating AI-Integrated Research Software: Demonstration and Validation through CyberFacCES Smart Search • A Nonprofit's Google Summer of Code Experience in the Age of AI-Assisted Contributions • Preparing Students for AI-Assisted Research Workflows Reflections on RSE Roles & Pathways , <i>San Jose Ballroom Salon 5 & 6</i> • What They Don't Tell You About the Data Science to RSE Transition • Bears, Bottlenecks, and Bus Factors: Why Embedded RSEs Pay Off • Mission Control: Service and the Stewardship of Science Building Infrastructure for Communities or Organizations , <i>Willow Glen Room</i> • From Lakehouse to Labmate: KBase's Vision for AI-Enabled Scientific Discovery • HEROIC: A Research Software Infrastructure for Real-Time Multi-Messenger Observatory Coordination • A Managed Database-as-a-Service Platform for Scientific Computing • One Size Fits None: Rethinking Campus Research Computing
3–3:15pm	Transition Break , <i>San Jose Marriott</i>
3:15–4:30pm	Rapid Access Microtalks (RAM), Closing Remarks , <i>San Jose Ballroom Salon 3 & 4</i>

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Keith Beattie, Lawrence Berkeley National Laboratory

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Talks: Adam Rubens (NYU Langone Health), Anas AlSobeh

Notebooks: Yuxi (Ethan) Cui, Primus Kabuo

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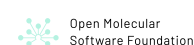
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